Meta-Analytic Evaluation of Volleyball Metrics

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Abstract

Evaluating the quality of metrics that assess player ability in sports helps stakeholders make effective comparisons between players without being overwhelmed by metrics that are heavily influenced by sampling variability or provide largely redundant information. In this paper, we apply such methods to evaluate the quality and underlying properties of a set of both conventional and advanced volleyball metrics on open-source professional women's volleyball data from Major League Volleyball, League One Volleyball Pro, and Athletes Unlimited Pro Volleyball. From these analyses, we identify and provide commentary on the metrics that provide the most unique and reliable information for the wider volleyball community, including coaches, athletes, and fans. By accounting for the heterogeneous distributions of league-specific data, we are able to compare and contrast the statistical properties of metrics across leagues. Such analysis has not been performed in the volleyball analytics literature and consequently fills a major gap, particularly with the recent proliferation of advanced metrics. Our findings provide a greater understanding of the properties of current statistics and identify areas where improvements can be made when developing new metrics.

1 Introduction

Quantitative analysis has become increasingly important in evaluating player performance and informing decision-making in sport. As richer data sources and more sophisticated statistical methods become available, the number and complexity of performance evaluation metrics have grown accordingly. Although this expansion provides analysts, coaches, and other stakeholders with more information, it also creates a fundamental problem: the existence of many metrics does not imply that each provides reliable or distinct information about performance. Consider volleyball, for example. Indoor volleyball is a team sport in which two teams of six players attempt to make the ball touch the opponent's court while preventing the opponent from doing the same. Players occupy specialized roles that constrain both their responsibilities and their opportunities to perform particular actions. Consequently, individual volleyball performance is multidimensional and role-dependent, such that a metric that is informative for one position may have limited relevance for another. Previous studies have used box-score-based statistics to assess the contributions of spike, serve, and block performance to team rankings (Marcelino et al. 2008), as well as identify which volleyball skills and contextual factors best predict winning and losing at the elite level (Peña et al. 2013). While providing aggregate insights, these works have not demonstrated how well these statistics distinguish between players or how closely they relate to one another. The volleyball analytics literature has recently focused on estimating individual contribution using increasingly sophisticated methods such as Markov chains and Bayesian logistic regression (Miskin et al. 2010, Powers et al. 2025). Complementary work has also examined subjective skill evaluation ratings (Bagley & Ware 2017), in-game serving strategies (Burton & Powers 2015), and setting tendencies (Raymond et al. 2024). Player impact estimates using hierarchical models have adapted approaches common to other sports (Hass & Craig 2018, Fellingham 2022), and the increasing availability of public data has expanded the prevalence of context-aware and teammate-agnostic metrics (Evollve 2026).

As advanced metrics become more commonplace, it is imperative to ask how their statistical properties can be properly evaluated. For example, a metric may have an intuitive interpretation or a sophisticated derivation while still being strongly affected by limited playing time or random variation, and several apparently different metrics may encode largely overlapping information. These issues are especially relevant when comparing players, where observed differences should ideally reflect systematic differences in performance rather than sampling variability. Meta-analytics provide a systematic way to study these properties without requiring all metrics to have the same scale, distribution, or underlying construction. In this paper, we adapt this framework to statistics used across different leagues in professional women's volleyball. Our objective is not to identify a single universally "best" volleyball metric, but rather to understand what information each metric provides and to identify where random variation or overlap between metrics may make the results harder to interpret.

1.1 Related Work

In their seminal paper, Franks et al. (2016) formalized their **meta-analytic** framework based on three criteria: *stability*, *discrimination*, and *independence*. Stability quantifies how consistent a player's value for a metric remains across seasons after accounting for random noise, capturing its sensitivity to change. Discrimination measures how well a metric differentiates between players based on the proportion of total between-player variance in a metric that reflects true differences in player ability rather than sampling variability. Finally, independence measures how much unique information a new metric adds beyond a set of previously considered metrics. They applied these definitions to evaluate more than 70 player-performance metrics from the National Basketball Association (NBA) and 40 metrics from the National Hockey League (NHL). Their analysis revealed substantial variation in the statistical properties of familiar performance measures. In the NBA, `rebounds`, `blocks`, and `assists` were highly discriminative and stable, whereas raw `three-point percentage` exhibited comparatively low discrimination and stability. The analysis also showed substantial overlap among several overall measures of player performance, with a principal component analysis of five such metrics showing that the first principal component explained approximately 73% of the variation across these measures. In the NHL, `hits`, `blocks`, and `shots` were relatively discriminative and stable, while `+/-` exhibited comparatively low discrimination; the framework also highlighted overlap among related families of measures such as `Corsi` and `Fenwick`. Meta-analytics have subsequently been applied to soccer statistics to examine how strongly metrics differentiate players and how their values persist over time (ElHabr 2023). More recently, Matteo et al. (2026) examined how well 33 player- and team-level performance indicators distinguished performance and how consistently they performed across nine seasons of Australian Football. They found that count-based measures such as `kicks`, `handballs`, `disposals`, and `goals` generally exhibited high discrimination and stability, whereas several percentage-based metrics were less reliable. These properties also varied across playing positions, reinforcing the importance of context when interpreting performance measures. To our knowledge, however, an analogous systematic evaluation has not been conducted for volleyball, which this work sets out to accomplish.

2 Data

We use open-source box-score and play-by-play data from the `pyvolleydata` Python library (Awosoga et al. 2026*a*) and its R equivalent `rvolleydata` (Awosoga et al. 2026*b*). Box-score records provide aggregated player statistics such as serving, attacking, reception, setting, digging, and blocking outcomes and support conventional counting, rate, percentage, and efficiency metrics. Player-information records additionally provide information necessary to identify performances by players across matches and seasons. Play-by-play data retain the sequence and context of actions within rallies, including information such as the set and point, team and player involved, action outcome, score, and point winner. These data support the development of more advanced metrics. The leagues considered in this study are Major League Volleyball (MLV), League One Volleyball Pro (LOVB Pro), and Athletes Unlimited Pro Volleyball (AU Pro), three major professional women's volleyball leagues in North America. We consider regular season games from MLV (308, 2024–2026), LOVB Pro (108, 2025 & 2026), and AU Pro (114, 2022–2025). AU Pro has a unique structure compared to MLV and LOVB Pro because there are no fixed teams and lineups are instead redrafted on a weekly basis during the season. The analysis therefore separates player identity from contextual team assignment. Additionally, "Golden Set" observations in AU Pro are excluded from the ordinary player-performance scope rather than treated as standard sets. Extensive data cleaning is performed on league-specific data, such as competition filtering, source corrections, team aliases, name disambiguation, and missing value treatment. Because the distribution and availability of metrics vary across competitions, all subsequent analyses are conducted separately within each league.

3 Methodology

Let X_{spm} denote the observed value of metric m for player p in season s, and let $V_{spm}[X]$ denote its sampling variability. We evaluate metrics according to discrimination, stability, and independence, with all meta-analytic quantities estimated separately within each league. The definitions and implementations of the individual metrics are described in Section 4.

3.1 Discrimination

Discrimination measures the extent to which observed differences between players within a season reflect systematic between-player variation rather than sampling noise. For metric m in season s,

$$\mathcal{D}_{sm} = 1 - \frac{E_{sm}\left[V_{spm}[X]\right]}{V_{sm}[X]}.$$

Values near one indicate that observed differences between players are large relative to estimated sampling variability, whereas lower values indicate a greater contribution from sampling noise. Discrimination is estimated separately for each season, and we also retain the across-season average

$$\mathcal{D}_m = E_m[\mathcal{D}_{sm}].$$

The primary discrimination value displayed alongside stability is the estimate from the most recent season available for each league.

3.2 Stability

Stability measures how consistently a player's metric value persists across seasons after accounting for sampling variability:

$$\mathcal{S}_m = 1 - \frac{E_m\left[V_{pm}[X] - V_{spm}[X]\right]}{V_m[X] - E_m\left[V_{spm}[X]\right]}.$$

Higher values indicate greater consistency across seasons after accounting for estimated sampling noise. We retain the same population-level quantity defined by Franks et al. (2016), but apply a small correction to its estimation to account for finite sample size. The original estimator calculates each player's within-player variance using

$$\frac{1}{S_p}\sum_{s=1}^{S_p}\left(X_{spm} - \bar{X}_{\cdot pm}\right)^2,$$

where S_p is the number of qualifying seasons for player p. In our data, many players are observed in only two or three qualifying seasons. With so few season-level observations per player, using the $1/S_p$ normalization tends to underestimate the within-player variance. We therefore use the standard unbiased sample-variance normalization,

$$\frac{1}{S_p - 1}\sum_{s=1}^{S_p}\left(X_{spm} - \bar{X}_{\cdot pm}\right)^2.$$

This adjustment changes only how the within-player variance is estimated for a finite number of seasons; it does not change the underlying quantity being measured. We thus use the corrected estimator as our primary stability measure, and only players with at least two valid qualifying seasons contribute to stability. Because the theoretical $[0, 1]$ bounds apply to the population quantity rather than necessarily to a finite-sample noise-corrected estimate, estimated stability values are not clipped when they fall outside this interval.

3.3 Independence

Independence measures how much unique information a metric provides compared to other metrics. Let C denote the latent correlation matrix among the metrics and $\mathcal{M}$ the conditioning set for target metric m:

$$\mathcal{I}_{m\mathcal{M}} = C_{m,m} - C_{m,\mathcal{M}} C_{\mathcal{M},\mathcal{M}}^{-1} C_{\mathcal{M},m}.$$

Equivalently, $\mathcal{I}_{m\mathcal{M}} = 1 - R^2$ on the latent Gaussian scale, so lower values indicate greater redundancy with the conditioning metrics and higher values indicate more unique information. We estimate C using the semiparametric rank-likelihood approach of Hoff (2007) within the `sbgcop` R package (Hoff 2012). Seasons are pooled within each league, metric-defined missing values are retained, and no complete-case or reliability-based filtering is applied. The resulting independence populations contain 392 MLV, 192 LOVB Pro, and 177 AU Pro player-seasons. Each league-specific model uses 5,000 MCMC iterations, retaining 1,000 posterior correlation matrices whose posterior mean is used for C. We then construct greedy independence curves, perform principal component analysis (PCA) on C, and use complete-linkage hierarchical clustering with Euclidean distances between the rows of $|C|$ to identify metrics with similar correlation profiles.

3.4 Estimation

Sampling variability for discrimination and stability is estimated using game-level bootstraps. Within each season, each team independently resamples its played matches with replacement, and all player observations belonging to a selected team-match are retained. For AU Pro, temporary week-squad assignments serve as the corresponding team blocks. The reliability population is fixed from the observed data before resampling, with a player-season required to contain at least 20 played sets and at least 20 recorded contacts related to the metric of interest. These eligibility thresholds are applied before resampling and are not reapplied within individual bootstrap replicates. We then generate 100 bootstrap replicates for each league-season. Metrics based on counts and rates are recalculated using their underlying data, while metrics requiring deterministic recalculation or model fitting are recomputed according to the procedures described in Section 4. For each qualifying player-season and metric, the variance of its finite bootstrap replicate values, $BV[X_{spm}]$, is used to estimate $V_{spm}[X]$. The remaining variance components required for discrimination and stability are estimated from the observed qualifying player-season values, producing final reliability populations of 321 MLV, 157 LOVB Pro, and 136 AU Pro player-seasons. For corrected stability, uncertainty is additionally summarized using a player-cluster bootstrap in which players are resampled as clusters so that all qualifying seasons belonging to a selected player remain together. The corrected stability estimator is recomputed for each draw, and percentile-based 95% confidence intervals are obtained from 2,000 such resamples.

4 Metrics

We evaluate fourteen conventional player-performance metrics derived from commonly reported volleyball box-score statistics (Major League Volleyball 2024). Ten are expressed on a per-set basis: `Kills/set (K/S)`, `Attack Errors/set (E/S)`, `Assists/set (AST/S)`, `Service Aces/set (SA/S)`, `Service Errors/set (SE/S)`, `Reception Errors/set (RE/S)`, `Digs/set (D/S)`, `Block Touches/set (BT/S)`, `Blocks/set (B/S)`, and `Points/set (PTS/S)`, where points are the sum of kills, service aces, and blocks. We additionally consider `Kill Percentage (K%)`, `Hitting Efficiency (Eff)`, `Passing Efficiency (Pass Eff)`, and `Serving Efficiency (Srv Eff)`, which normalize performance by the corresponding number of attacking, reception, or serving opportunities. In addition to the conventional statistics, we evaluate three metrics from the volleyball literature, implemented as closely as possible based on available resources.

`Point Scoring Fraction (PSF)` was proposed by Burton & Powers (2015) to evaluate serving performance while jointly accounting for aces, service errors, and the opponent's ability to side out when the serve remains in play. It is defined as

$$\texttt{PSF} = a + (1 - a - e)(1 - s),$$

where a and e are the proportions of serves resulting in aces and errors, respectively, and s is the opponent sideout proportion conditional on serves that end in neither an ace nor an error.

`Attack Evenness (AEV)` measures how evenly a setter distributes attacking opportunities among available hitters relative to a reference attack distribution (Raymond et al. 2024). Given an observed attack distribution $\mathbf{a}$ and reference distribution $\mathbf{r}$,

$$\texttt{AEV} = 1 - \frac{1}{2} \sum_i |a_i - r_i|,$$

with larger values indicating a distribution closer to the reference profile. We calculate `AEV` using the authors' `ovlytics` R package (Raymond & Ickowicz 2024), reconstructing the required lineup and setter information and preserving the source's attack-volume and reporting requirements.

`Adjusted Plus-Minus (APM)` estimates a player's association with point outcomes while simultaneously accounting for their teammates and opponents (Hass & Craig 2018). Player effects are estimated through logistic ridge regression and transformed as

$$50\,\mathrm{logit}^{-1}(\beta_j) - 25,$$

where β_j is the fitted coefficient for player j. We reconstruct the model using complete observed lineups and complementary observations of each point from the two teams' perspectives. We select a penalty term using 10-fold cross-validation.

Finally, we evaluate thirteen advanced metrics defined in the public Evollve statistics glossary (Evollve 2026). `Serve-Receive Hitting Percentage (SR Hit %)` measures attacking performance on attacks occurring during the serve-receive phase and is calculated as

$$\frac{\text{serve-receive attack kills} - \text{serve-receive attack errors}}{\text{serve-receive attack opportunities}}.$$

`Transition Hitting Percentage (Trans Hit %)` applies analogously to attacks occurring after a dig:

$$\frac{\text{transition attack kills} - \text{transition attack errors}}{\text{transition attack opportunities}}.$$

`Serve-Receive Hit Average (SR Hit Avg)` is a variant of `Serve-Receive Hitting Percentage (SR Hit %)` that assigns partial credit to non-terminal attacks during serve receive:

$$\frac{\text{serve-receive kills} + 0.433(\text{serve-receive non-terminal attacks})}{\text{serve-receive attack opportunities}}.$$

The coefficient of 0.433 is retained from the reference implementation. `Transition Hit Average (Trans Hit Avg)` applies the corresponding weighted construction to transition attacks:

$$\frac{\text{transition kills} + 0.433(\text{transition non-terminal attacks})}{\text{transition attack opportunities}}.$$

`Dig Attack Conversion Percentage (Dig→Att %)` measures how often a successful dig is followed by a subsequent attack by the same team. `Dig Kill Percentage (Dig→Kill %)` measures how often a successful dig is followed by a subsequent attack that results in a kill. `Dig Rate` measures the proportion of opponent attacking opportunities for which an on-court player records a dig:

$$\frac{\text{player digs}}{\text{opponent attacks while the player is on court}},$$

with on-court exposure calculated only where complete lineup information is available.

`Reception Rate (Rec Rate)` measures the proportion of opponent serve opportunities received by a player while that player is on court:

$$\frac{\text{player receptions}}{\text{opponent serves while the player is on court}}.$$

`Percentage of Points Won While On Court (% Won On)` measures the proportion of points won by a player's team while that player is on court:

$$\frac{\text{team points won while the player is on court}}{\text{points played while the player is on court}}.$$

`Kill Rate per Rally Point (K/Rally)` expresses a player's kills relative to the number of rally points for which that player is on court:

$$\frac{\text{kills}}{\text{on-court rally points}},$$

where rally points exclude points terminating directly on a service ace or service error.

`Blocked Rate (Blocked %)` measures the proportion of a player's attack attempts that result in a blocked attack error. `Serve Average (Srv Avg)` assigns full credit to aces, zero credit to service errors, and partial credit to other serves:

$$\frac{\text{aces} + 0.435(\text{serves} - \text{aces} - \text{service errors})}{\text{serves}},$$

using the published weight of 0.435. Finally, `Attack Error Rate Excluding Blocks (Att Err %)` measures attack errors that are not attributed to the attacker being blocked:

$$\frac{\text{unblocked attack errors}}{\text{attack attempts}}.$$

Because only three of the thirteen Evollve metrics can be computed directly from the available AU Pro box-score data, only `Blocked Rate (Blocked %)`, `Serve Average (Srv Avg)`, and `Attack Error Rate Excluding Blocks (Att Err %)` are evaluated for that league.

5 Results

5.1 Discrimination and Stability

The left panels of Figure 1–Figure 3 compare latest-season discrimination with corrected stability across the three leagues. Several conventional per-set measures were consistently highly discriminative across leagues: `Assists/set` had the largest estimate in MLV (0.991), LOVB Pro (0.989), and AU Pro (0.993), while `Kills/set` and `Points/set` also exceeded 0.96 in each league. In MLV and LOVB Pro, `Reception Rate` was similarly discriminative (0.951 in both), whereas several contextual advanced metrics showed substantially lower values, including `Dig Kill Percentage` in MLV (0.147) and LOVB Pro (0.049) and `Dig Attack Conversion Percentage` in LOVB Pro (0.112). AU Pro exhibited a narrower range, with `Passing Efficiency` producing its lowest discrimination estimate (0.481).

Corrected stability varied more strongly across leagues. MLV produced high estimates for `PSF` (1.660), `Hitting Efficiency` (1.151), and `Kill Percentage` (1.076), while `AEV` (0.536), `Serve Average` (0.338), and `Blocked Rate` (-0.091) were less stable. In LOVB Pro, `Dig Attack Conversion Percentage` (1.166) and `Transition Hit Average` (1.150) had high estimates, whereas several metrics produced negative estimates, including `Attack Error Rate Excluding Blocks` (-0.470), `Serve-Receive Hitting Percentage` (-0.298), and `PSF` (-0.256). AU Pro estimates remained within the unit interval, with `Assists/set` (0.971), `Blocks/set` (0.931), and `Digs/set` (0.919) among the most stable and `Hitting Efficiency` (0.199) and `Kill Percentage` (0.386) among the least stable. Several corrected stability estimates for MLV and LOVB Pro were below zero or above one because the finite-sample correction can produce values outside the theoretical range, and these values were kept unchanged rather than capped at zero or one. `Dig Kill Percentage` did not produce a corrected stability estimate for LOVB Pro because the adjusted between-player variation was zero or negative. As these leagues continue to operate in more seasons, we expect stability values to more readily fall within the $[0, 1]$ interval.

5.2 Independence

The right panels of Figure 1–Figure 3 present the greedy conditional-independence curves for the three leagues. Under full conditioning, `Passing Efficiency` was among the most independent metrics in all three leagues, with values of 0.787 in MLV, 0.858 in LOVB Pro, and 0.795 in AU Pro. In MLV and LOVB Pro, `AEV` and the two dig-conversion measures also provided relatively unique information: `AEV` reached 0.739 and 0.890, respectively, while `Dig Kill Percentage` reached 0.778 and 0.866. Several commonly reported metrics were substantially more redundant, with `Points/set` and `Kills/set` among the lowest full-conditioning independence measures in both MLV and LOVB Pro. The serving family, including `Serve Average` and `Serving Efficiency`, also showed low independence across leagues. In AU Pro, `Service Errors/set` had the lowest independence (0.105), followed by `Service Aces/set` (0.166), `Serving Efficiency` (0.171), and `Serve Average` (0.177).

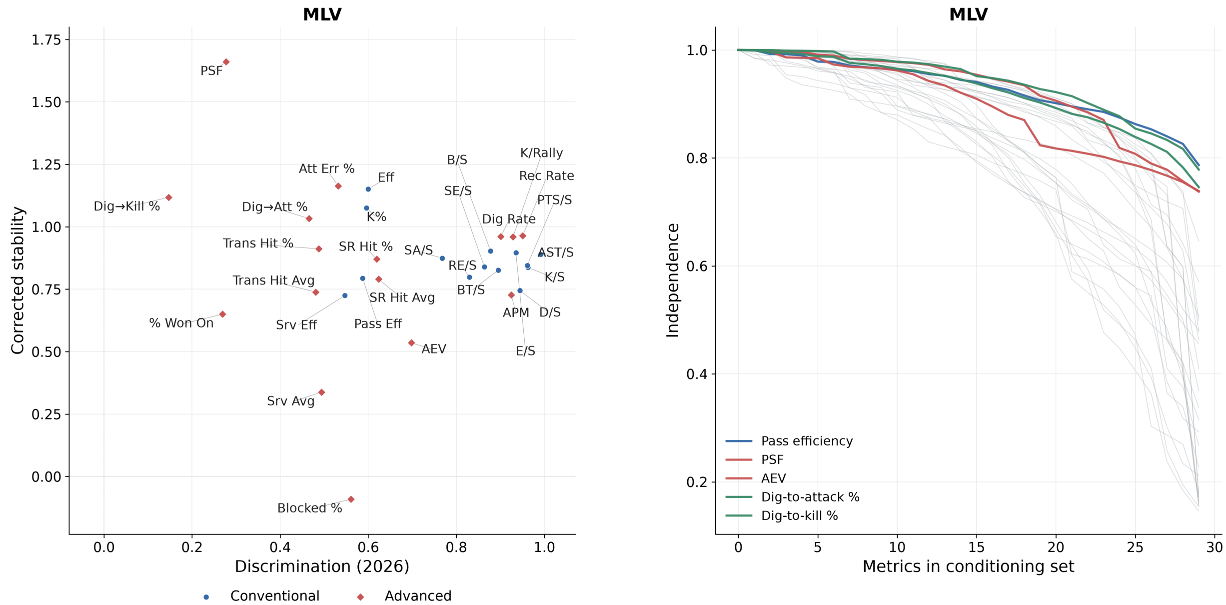

Figure 1: Meta-analytic results for MLV. The left panel shows latest-season discrimination and corrected stability, with estimates shown as calculated, including values outside the theoretical range. The right panel shows the greedy conditional-independence curves, with the five metrics having the largest full-conditioning independence values highlighted.

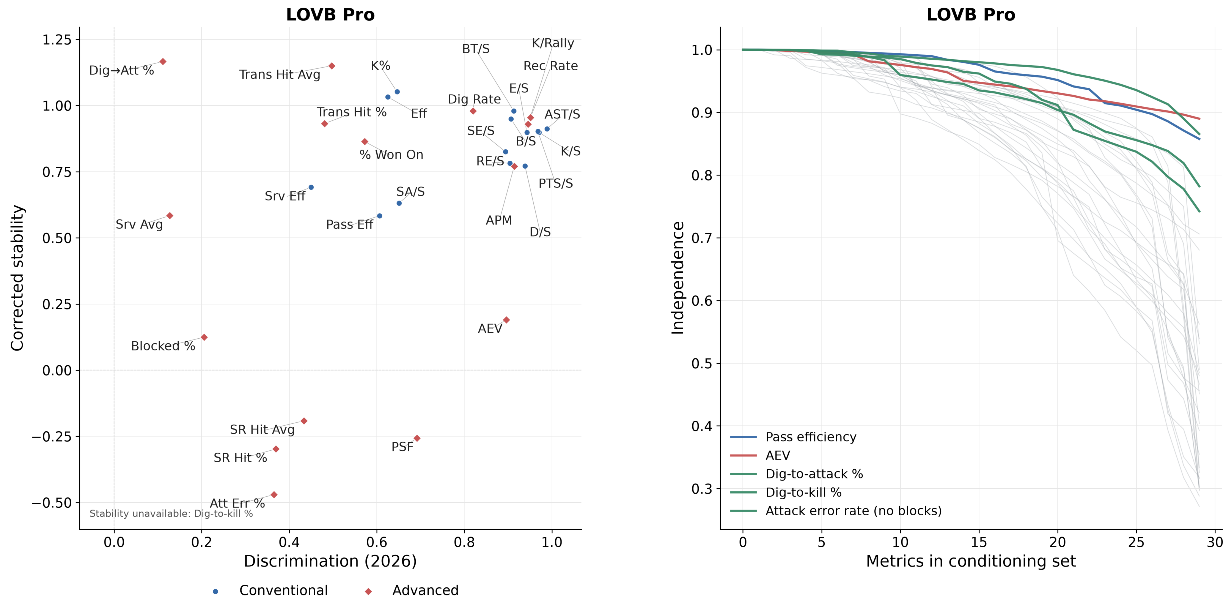

Figure 2: Meta-analytic results for LOVB Pro. The left panel shows latest-season discrimination and corrected stability, with estimates shown as calculated, including values outside the theoretical range. The right panel shows the greedy conditional-independence curves, with the five metrics having the largest full-conditioning independence values highlighted.

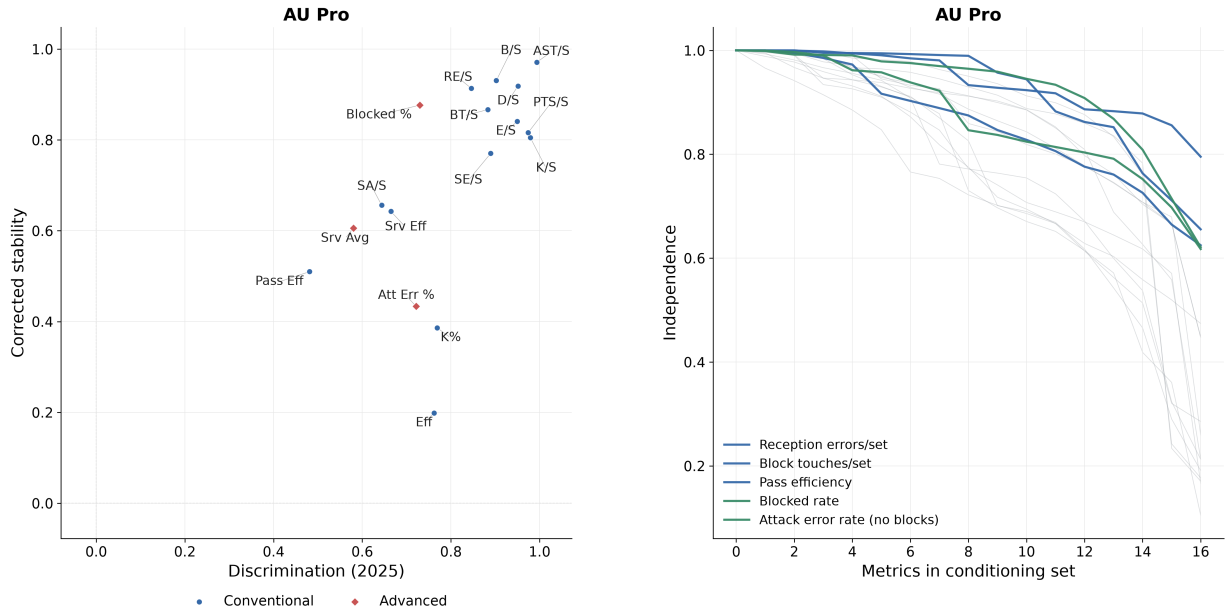

Figure 3: Meta-analytic results for AU Pro. The left panel shows latest-season discrimination and corrected stability. The right panel shows the greedy conditional-independence curves, with the five metrics having the largest full-conditioning independence values highlighted.

5.3 PCA and Clustering

Principal component analysis of the latent correlation matrices indicated substantial but incomplete redundancy among the analyzed metrics. As shown in Figure 4, MLV required 11, 16, and 21 principal components to explain 80%, 90%, and 95% of latent variance, respectively. LOVB Pro required 13, 19, and 23 components, while AU Pro required 7, 10, and 12 components from its smaller 17-metric set.

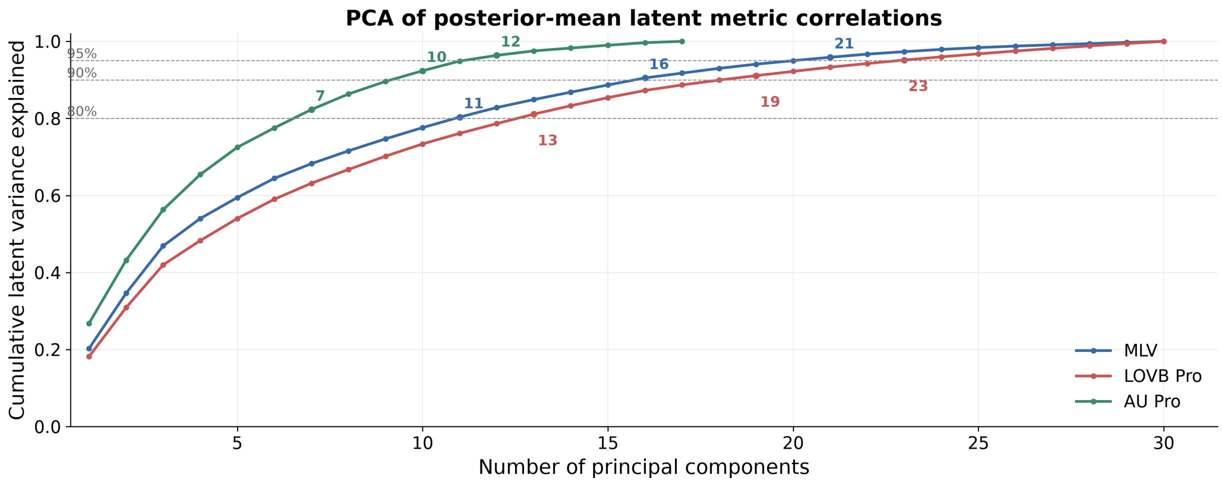

Figure 4: Cumulative variance explained by principal components of the posterior-mean latent correlation matrix for MLV, LOVB Pro, and AU Pro.

The hierarchical clustering results in Figure 5–Figure 7 further characterize similarities in metric correlation profiles. `Kills/set` and `Points/set` clustered closely in all three leagues, as did `Serve Average` and `Serving Efficiency`. In MLV and LOVB Pro, `Serve-Receive Hit Average` clustered closely with `Serve-Receive Hitting Percentage`, while `Transition Hit Average` clustered with `Transition Hitting Percentage`. These clusters reflect similarity across rows of the absolute latent correlation matrix rather than only pairwise correlation between the metrics themselves.

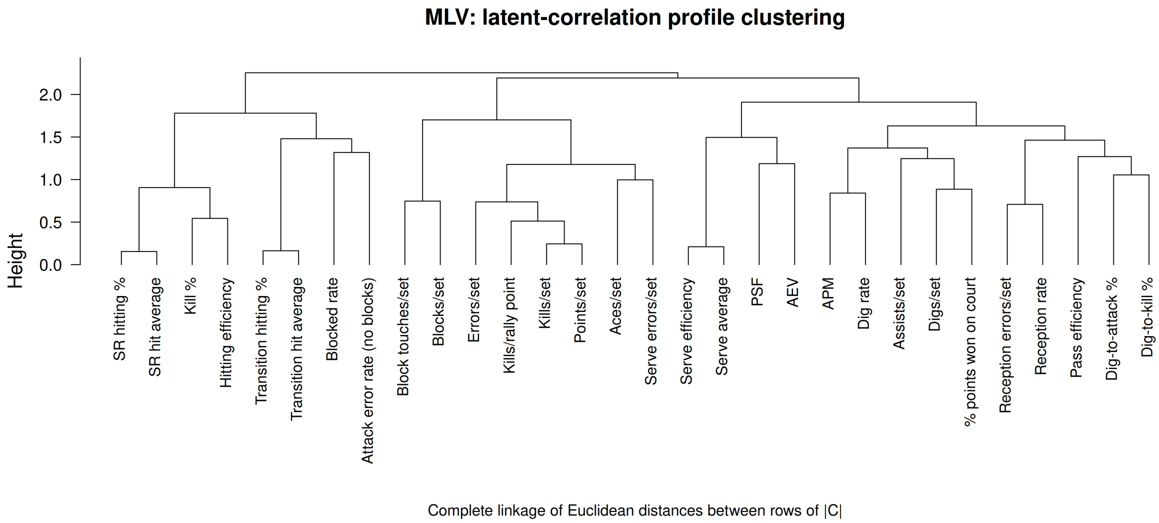

Figure 5: Complete-linkage hierarchical clustering of metric correlation profiles for MLV. Distances are Euclidean distances between rows of the absolute posterior-mean latent correlation matrix.

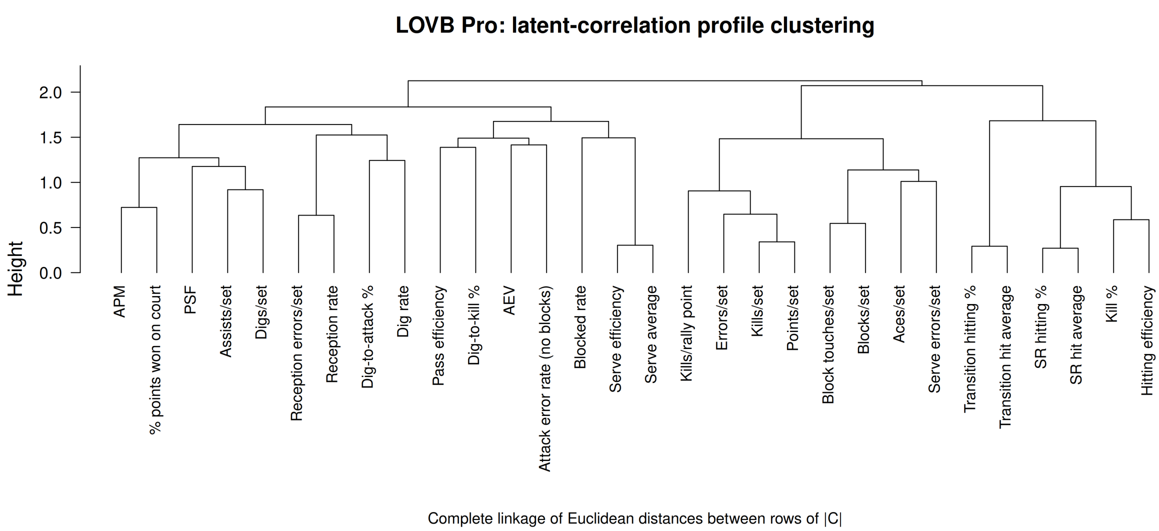

Figure 6: Complete-linkage hierarchical clustering of metric correlation profiles for LOVB Pro. Distances are Euclidean distances between rows of the absolute posterior-mean latent correlation matrix.

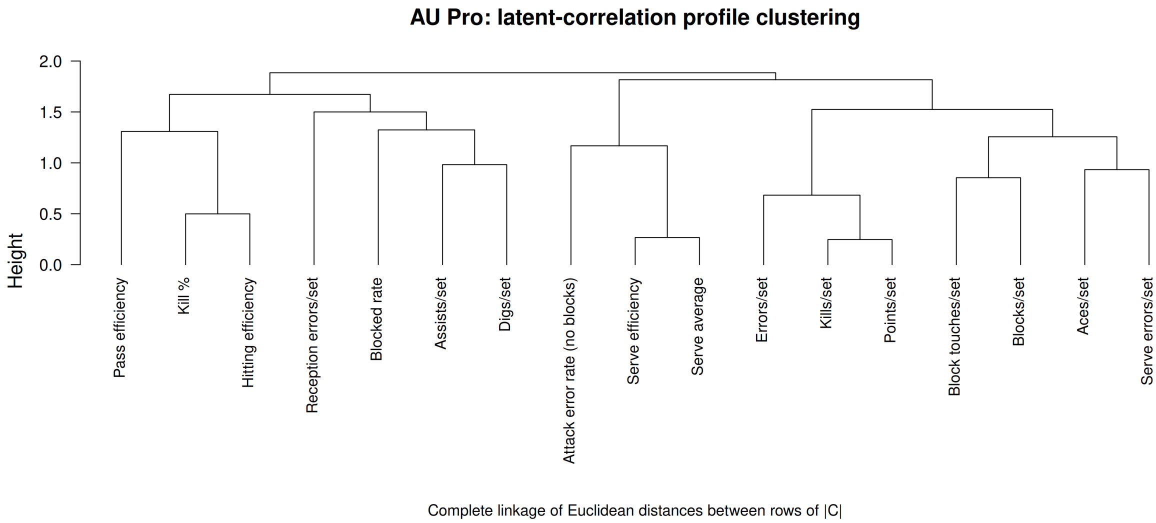

Figure 7: Complete-linkage hierarchical clustering of metric correlation profiles for AU Pro. Distances are Euclidean distances between rows of the absolute posterior-mean latent correlation matrix.

6 Discussion

The results demonstrate that discrimination, stability, and independence capture distinct properties of volleyball metrics and should therefore be considered jointly. Several conventional per-set statistics, particularly `Assists/set`, `Kills/set`, and `Points/set`, were highly discriminative across leagues but were also among the least independent metrics. Conversely, advanced metrics such as `AEV` and the dig-conversion measures contributed comparatively distinct information despite lower discrimination or stability in some settings. A metric can therefore distinguish players clearly without adding much new information, or provide comparatively nonredundant information while remaining sensitive to sampling variability or season-to-season change. The cross-league results further demonstrate that these statistical properties are not necessarily intrinsic to the metric itself: although some discrimination patterns were consistent, stability differed considerably across MLV, LOVB Pro, and AU Pro. For example, `PSF` had a large corrected stability estimate in MLV but a negative estimate in LOVB Pro. Such differences may reflect variation in player populations, season length, player movement, competitive structure, opportunity distributions, or available data, and a metric that behaves reliably in one competition should not automatically be assumed to have the same properties in another.

PCA and clustering reinforce the distinction between redundancy and unique information. Although the original set of metrics could be reduced, several principal components were still needed to capture most of the variation, suggesting that the metrics reflected more than just a few common underlying patterns. At the same time, intuitive clusters such as `Kills/set` with `Points/set` and `Serve Average` with `Serving Efficiency` identify groups of statistics that may provide overlapping descriptions of player performance. These properties have direct applications when deciding which statistics to communicate to practitioners. In a scouting report, discrimination and stability can help identify metrics that provide comparatively reliable separation between players, while independence can help avoid presenting several statistics that largely repeat the same underlying information. A report could therefore prioritize a smaller collection of complementary metrics rather than simply including every available statistic. Similarly, coaches could use reliable skill-specific measures to monitor areas of player performance that have already been identified as priorities. Meta-analytic evaluation does not tell coaches which training plan to use, but it can help determine whether the metric used to guide or monitor that plan is reliable and informative enough to interpret meaningfully.

7 Conclusion

This study applies a meta-analytic framework to evaluate the statistical properties of conventional and advanced player-performance metrics across MLV, LOVB Pro, and AU Pro. The results demonstrate that discrimination, stability, and independence provide complementary perspectives on metric quality, with no single property sufficient to characterize a metric on its own. Several conventional per-set statistics clearly distinguished between players but also contained substantial overlap with other measures, while some advanced metrics provided comparatively nonredundant information despite weaker discrimination or stability. The PCA and clustering analyses similarly showed that the analyzed metrics contain meaningful redundancy without collapsing into only a small number of common dimensions. Differences across leagues further demonstrate that the statistical behaviour of a metric depends in part on the competition, available data, and player population in which it is evaluated. Together, these findings show that the growing number of volleyball metrics should not be interpreted solely according to their definitions or apparent complexity, but also according to the statistical information they provide. For practitioners, meta-analytic evaluation can help support the selection of smaller and more complementary sets of statistics for scouting, player development, and performance communication. More broadly, this work provides a systematic foundation for evaluating existing and future volleyball metrics according to whether observed player differences exceed sampling noise, persist across seasons, and provide information beyond that already captured by other measures.

7.1 Limitations and Future Work

Several limitations should be considered when interpreting these results. Only a small number of seasons are available for each professional competition, leaving many repeated players with two or three qualifying seasons and limiting the precision with which stability can be estimated. Advanced measures requiring detailed sequence or lineup information also have more restricted coverage, and `AEV` in particular is available for relatively few player-seasons because of its source-defined setter and attack-volume requirements. Ambiguous sequence or lineup information is treated as missing rather than inferred, prioritizing source fidelity at the cost of coverage, while metric availability also differs across leagues, especially for AU Pro. Player specialization and differences in opportunity may additionally contribute to some of the observed statistical properties because players in different roles are exposed to substantially different actions and responsibilities.

Future work should first revisit these analyses as longer professional league histories and richer play-by-play and lineup data become available. Larger repeated-player populations would allow stability to be estimated more precisely, while improved data coverage would support more consistent evaluation of advanced metrics across leagues. A second direction is to expand the set of source-faithful metrics included in the analysis as additional measures become available or can be reliably reconstructed from public data. Evaluating a broader set of metrics would provide a more complete picture of which aspects of volleyball performance are already well represented and where meaningful gaps remain. The identified gaps can also inform the development of new volleyball metrics. In particular, new measures could be designed to provide information that is less redundant with existing statistics while also aiming for stronger discrimination and stability, and then evaluated using the same meta-analytic framework. Position-specific meta-analysis represents another important extension because volleyball roles differ substantially in both responsibilities and statistical opportunities. Evaluating setters, hitters, middle blockers, liberos, and other positional groups separately could help distinguish properties of the metrics themselves from patterns driven primarily by role and opportunity. The framework could also be extended beyond professional volleyball to collegiate competitions such as NCAA and U Sports volleyball. These competitions would provide larger and structurally different player populations in which to assess whether the patterns observed across MLV, LOVB Pro, and AU Pro generalize to other levels of play. Finally, meta-analytic properties could be considered alongside predictive or decision-specific objectives. Discrimination, stability, and independence do not directly measure predictive value or causal contribution to winning, so future work could separately evaluate how metrics with different meta-analytic profiles relate to outcomes such as point, set, or match success and to specific scouting or coaching decisions.

References

Awosoga, D., Chow, M. & Du, R. (2026*a*), 'pyvolleydata: Extract Data from Professional Volleyball Leagues in North America'.
URL: *https://github.com/ryanndu/pyvolleydata*

Awosoga, D., Chow, M. & Du, R. (2026*b*), 'rvolleydata: Extract Data from Professional Volleyball Leagues in North America'.
URL: *https://github.com/awosoga/rvolleydata*

Bagley, C. & Ware, B. (2017), 'Bump, Set, Spike: Using Analytics to Rate Volleyball Teams and Players'.

Burton, T. & Powers, S. (2015), 'A linear model for estimating optimal service error fraction in volleyball', *Journal of Quantitative Analysis in Sports* **11**(2), 117–129.
URL: *https://www.degruyterbrill.com/document/doi/10.1515/jqas-2014-0087/html*

ElHabr, T. (2023), 'Tony's Blog - Meta-Analytics for Soccer'.

URL: *https://tonyelhabr.rbind.io/posts/soccer-meta-analytics/*

Evollve (2026), 'Glossary | Evollve | NCAA Women's Volleyball Statistics'.

URL: *https://evollve.net/stats/glossary*

Fellingham, G. W. (2022), 'Evaluating the performance of elite level volleyball players', *Journal of Quantitative Analysis in Sports* **18**(1), 15–34.

URL: *https://www.degruyter.com/document/doi/10.1515/jqas-2021-0056/html?lang=en*

Franks, A. M., D'Amour, A., Cervone, D. & Bornn, L. (2016), 'Meta-analytics: tools for understanding the statistical properties of sports metrics', *Journal of Quantitative Analysis in Sports* **12**(4), 151–165.

URL: *https://www.degruyter.com/document/doi/10.1515/jqas-2016-0098/html?lang=en*

Hass, Z. & Craig, B. (2018), 'Exploring the potential of the plus/minus in NCAA women's volleyball via the recovery of court presence information', *Journal of Sports Analytics* **4**, 285–295.

Hoff, P. (2012), 'sbgcop: Semiparametric Bayesian Gaussian Copula Estimation and Imputation'.

URL: *https://CRAN.R-project.org/package=sbgcop*

Hoff, P. D. (2007), 'Extending the rank likelihood for semiparametric copula estimation', *The Annals of Applied Statistics* **1**(1).

URL: *https://doi.org/10.1214/07-AOAS107*

Major League Volleyball (2024), 'Statistics Guide and Definitions'.

URL: *https://provolleyball.com/statistics-guide*

Marcelino, R., Mesquita, I. & Afonso, J. (2008), 'The weight of terminal actions in Volleyball. Contributions of the spike, serve and block for the teams' rankings in the World League 2005', *International Journal of Performance Analysis in Sport* **8**(2), 1–7.
URL: *https://www.tandfonline.com/doi/full/10.1080/24748668.2008.11868430*

Matteo, D., Varley, M. C., Ruddy, J. D. & Carey, D. L. (2026), 'Understanding the statistical properties of performance indicators in Australian Football', *Journal of Sports Sciences* **44**(14), 1867–1876.
URL: *https://www.tandfonline.com/doi/full/10.1080/02640414.2026.2703355*

Miskin, M. A., Fellingham, G. W. & Florence, L. W. (2010), 'Skill Importance in Women's Volleyball', *Journal of Quantitative Analysis in Sports* **6**(2).
URL: *https://doi.org/10.2202/1559-0410.1234*

Peña, J., Rodríguez-Guerra, J., Buscà, B. & Serra, N. (2013), 'Which Skills and Factors Better Predict Winning and Losing in High-Level Men's Volleyball?', *Journal of Strength and Conditioning Research* **27**(9), 2487–2493.
URL: *https://journals.lww.com/00124278-201309000-00017*

Powers, S., Stancil, L. & Consiglio, N. (2025), 'Estimating individual contributions to team success in women's college volleyball', *Journal of Quantitative Analysis in Sports* **21**(2), 117–135.
URL: *https://www.degruyter.com/document/doi/10.1515/jqas-2024-0038/html*

Raymond, B. & Ickowicz, A. (2024), 'ovlytics: Functions and Algorithms for Volleyball Analytics'.
URL: *https://ovlytics.openvolley.org*

Raymond, B., Ickowicz, A., Lebedew, M. & Mattes, M. (2024), 'Attack evenness'.

URL: *https://untan.gl/attack-evenness.html*